Addressing non-convergence in Structural Equation Models with the latent R package

SEM WG 2026 Meeting

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The latent R package

- Core functions written in C++
- Highly customizable models
- lavaan syntax for Structural Equation Models

Console | Terminal | Background Jobs

R 4.5.1 · ~/latent-master/

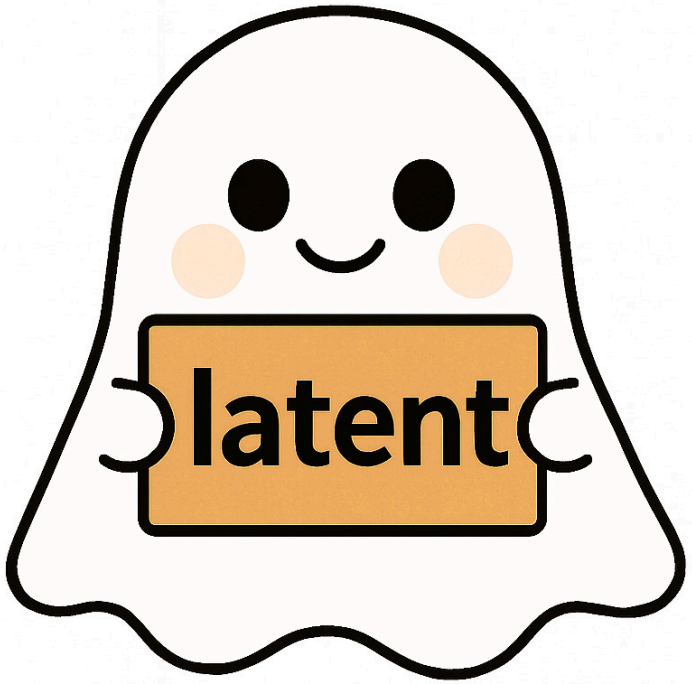
```
> library(latent)

Welcome to latent!

     __       __              __
    / /_____/ /____  ____  / /_
   / / __  / __/ _ \/ __ \/ __/
  / / /_/ / /_/  __/ / / / /_
 /_/\____/\__/\___/_/ /_/\__/   version 0.1.0

Type 'citation("latent")' for citing this package in publications.

Report bugs at m.j.jimenezhenriquez@vu.nl
For tutorials, visit marcosjnez.github.io/latent/
```



Example: Confirmatory Factor Analysis Example

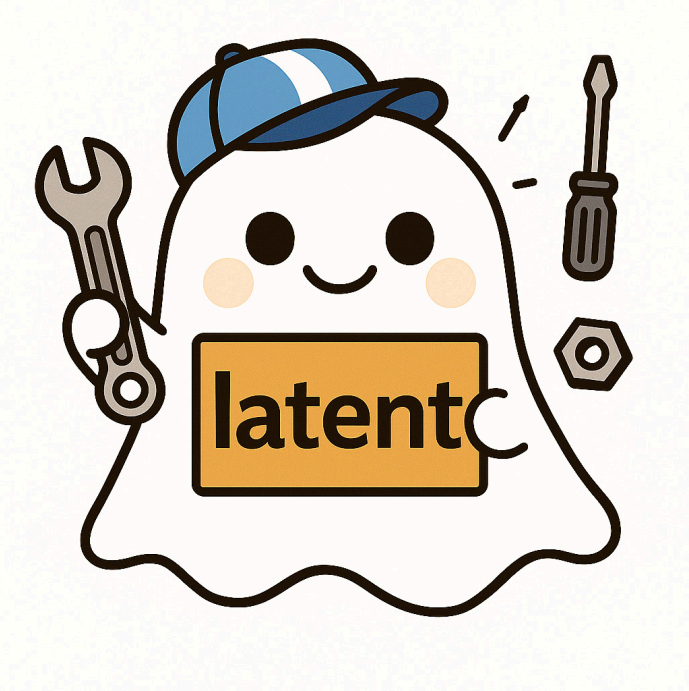
Model fitting:

```r
HS.model <- ' visual  =~ x1 + x2 + x3
              textual =~ x4 + x5 + x6
              speed   =~ x7 + x8 + x9 '
fit <- lcfa(model = HS.model, data = HolzingerSwineford1939)
```

Extract model information:

```r
# Get fit indices:
lavaan::fitMeasures(fit)

# Inspect model objects:
lavaan::inspect(fit, what = "est", digits = 3) # Estimates
lavaan::inspect(fit, what = "se", digits = 3)  # Standard errors
```



Problem: Positive-definite constraints

In the factor model equation,

$$S = \Lambda \Psi \Lambda^\top + \Theta,$$

Latent correlations Ψ and covariances Θ should be positive-definite but...

```
Warning message:
lavaan->lav_object_post_check():
   covariance matrix of latent variables is not
   positive definite ; use lavInspect(fit, "cov.lv")
   to investigate.
```

Positive-definite constraints (`lavaan` fails)

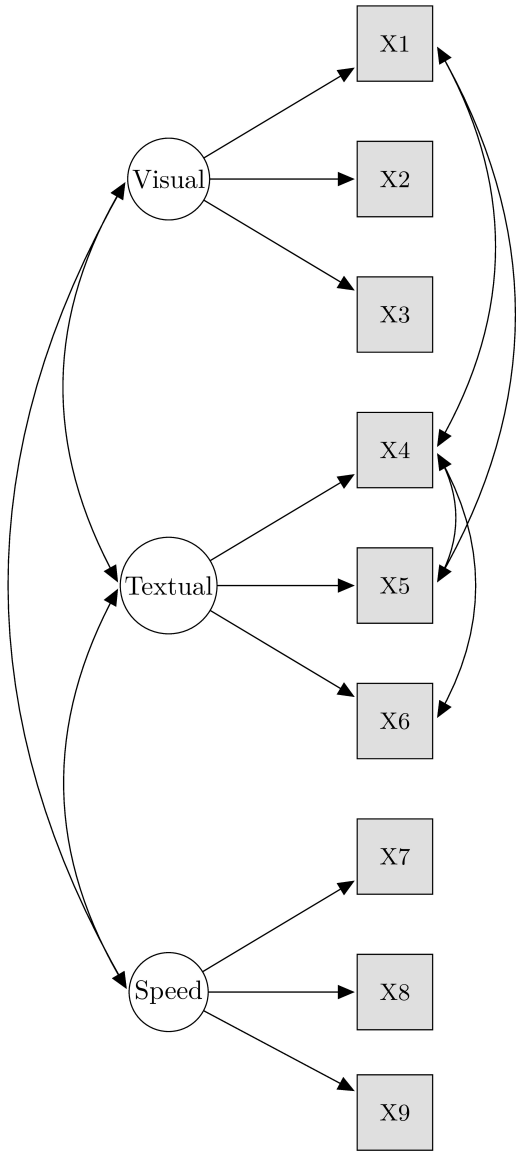
Let's force lavaan to fail for convergence:

```r
library(lavaan)
model <- 'visual  =~ x1 + x2 + x3
          textual =~ x4 + x5 + x6
          speed   =~ x7 + x8 + x9
          x1 ~~ x5
          x1 ~~ x4
          x4 ~~ x5
          x4 ~~ x6'
fit <- cfa(data = HolzingerSwineford1939, model = model, estimator = "ml")
inspect(fit, what = "est")$theta      # Error covariances
```

```
      x1     x2     x3     x4     x5     x6     x7     x8     x9
x1  0.621
x2  0.000  1.117
x3  0.000  0.000  0.791
x4  0.115  0.000  0.000  0.331
x5  0.090  0.000  0.000  0.199  0.867
x6  0.000  0.000  0.000 -0.258  0.000 -0.105
x7  0.000  0.000  0.000  0.000  0.000  0.000  0.792
x8  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.467
x9  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.586
```

```r
det(inspect(fit, what = "est")$theta) # Check the determinant
```

```
[1] -0.009621445
```



Positive-definite constraints (`latent` converges)

```r
fit <- lcfa(data = HolzingerSwineford1939, model = model,
            estimator = "ml", positive = TRUE)
round(latInspect(fit, what = "est")[[1]]$theta, 3) # Error covariances
```

```
      x1    x2    x3     x4    x5     x6    x7    x8    x9
x1 0.616 0.000 0.000  0.115 0.067  0.000 0.000 0.000 0.000
x2 0.000 1.115 0.000  0.000 0.000  0.000 0.000 0.000 0.000
x3 0.000 0.000 0.798  0.000 0.000  0.000 0.000 0.000 0.000
x4 0.115 0.000 0.000  0.461 0.201 -0.111 0.000 0.000 0.000
x5 0.067 0.000 0.000  0.201 0.755  0.000 0.000 0.000 0.000
x6 0.000 0.000 0.000 -0.111 0.000  0.061 0.000 0.000 0.000
x7 0.000 0.000 0.000  0.000 0.000  0.000 0.795 0.000 0.000
x8 0.000 0.000 0.000  0.000 0.000  0.000 0.000 0.471 0.000
x9 0.000 0.000 0.000  0.000 0.000  0.000 0.000 0.000 0.581
```

```r
det(latInspect(fit, what = "est")[[1]]$theta) # Check the determinant
```

```
[1] 0.001039987
```

Positive-semidefinite latent covariances

- Parameterize latent covariances as crossproducts:

$$\Psi = Y^\top Y$$
$$\Theta = U^\top U$$

- Constraint **Y** and **U** to be **orthoblique**:

$$X \in \mathbb{R}^{p \times p} : X^\top X = \text{sparse matrix}$$

$$\begin{bmatrix} 0.08 & 1.76 & 0.04 \\ -1.95 & -0.12 & -0.69 \\ -0.67 & -0.08 & 2.02 \end{bmatrix}^\top \begin{bmatrix} 0.08 & 1.76 & 0.04 \\ -1.95 & -0.12 & -0.69 \\ -0.67 & -0.08 & 2.02 \end{bmatrix} = \begin{bmatrix} 4.24 & 0.42 & 0.00 \\ 0.42 & 3.11 & 0.00 \\ 0.00 & 0.00 & 4.56 \end{bmatrix}$$

Positive-semidefinite latent correlations

- Parameterize latent covariances as crossproducts:

$$\Psi = Y^\top Y$$
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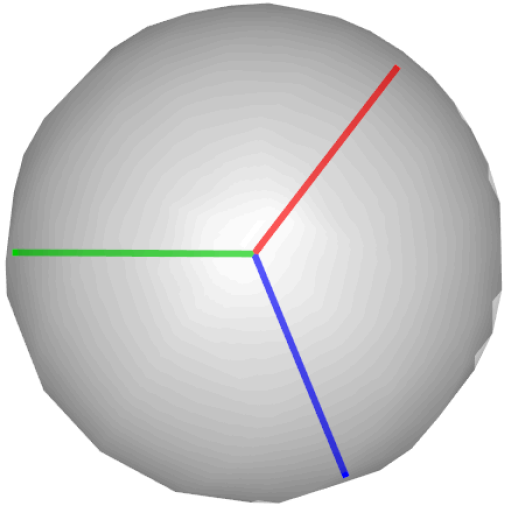

- Constraint **Y** and **U** to be **orthoblique**:

$$X \in \mathbb{R}^{p \times p} : X^\top X = \text{sparse matrix}$$

$$\begin{bmatrix} 0.04 & 1.00 & 0.02 \\ -0.95 & -0.07 & -0.32 \\ -0.32 & -0.05 & 0.95 \end{bmatrix}^\top \begin{bmatrix} 0.04 & 1.00 & 0.02 \\ -0.95 & -0.07 & -0.32 \\ -0.32 & -0.05 & 0.95 \end{bmatrix} = \begin{bmatrix} 1.00 & 0.12 & 0.00 \\ 0.12 & 1.00 & 0.00 \\ 0.00 & 0.00 & 1.00 \end{bmatrix}$$

This is like factor rotation in EFA

$$S = \underbrace{\Lambda X^{-\top}}_{\Lambda^*} \underbrace{(X^\top X)}_{\Psi} \underbrace{(\Lambda X^{-\top})^\top}_{\Lambda^{*\top}} + \Theta$$

$$= \Lambda^* \Psi \Lambda^{*\top} + \Theta.$$

Type of constraint

- **Orthogonal rotation**:
  $X \in \mathbb{R}^{p \times p}$ such that $X^\top X = I$.
- **Oblique rotation**:
  $X \in \mathbb{R}^{p \times p}$ such that $X^\top X = \text{dense matrix (with unit diagonal)}$.
- **Orthoblique rotation**:
  $X \in \mathbb{R}^{p \times p}$ such that $X^\top X = \text{sparse matrix (with unit diagonal)}$.

This is like factor rotation in EFA

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New factor rotation in EFA

PSYCHOMETRIKA—VOL. 66, NO. 2, 289–306
JUNE 2001

A SIMPLE GENERAL PROCEDURE FOR ORTHOGONAL ROTATION

ROBERT I. JENNRICH
UNIVERSITY OF CALIFORNIA AT LOS ANGELES

PSYCHOMETRIKA—VOL. 67, NO. 1, 7–20
MARCH 2002

A SIMPLE GENERAL METHOD FOR OBLIQUE ROTATION

ROBERT I. JENNRICH
UNIVERSITY OF CALIFORNIA AT LOS ANGELES

The orthoblique rotation for exploratory factor analysis

Marcos Jiménez Mauricio Garnier-Villarreal
Vithor R. Franco

From positive-semidefinite to positive-definite

We may maximize the log likelihood subject to a weighted penalty

$$\underset{\theta}{\operatorname{argmax}} \;\; \ell(\theta) + w \cdot \text{penalty}$$

Suppose the penalty only involves Ψ.

- The parameterization $\Psi = Y^\top Y$ is an square, so $\det \Psi \geq 0$.
- Notice that $0 \leq \det(D^{-1/2}\Psi D^{-1/2}) \geq 1$, where $D = \operatorname{diag}(\Psi)$.

Therefore, a good penalty term for Ψ becomes

$$\log \det(D^{-1/2}\Psi D^{-1/2}).$$

Summary

In the factor model $S = \Lambda\Psi\Lambda^\top + \Theta$,

1. the following parameterization will ensure, at least, positive-semidefinite matrices:

$$\Psi = Y^\top Y$$
$$\Theta = U^\top U,$$

   where Y and U are **orthoblique** matrices (i.e., $X \in \mathbb{R}^{p\times p} : X^\top X = \text{sparse matrix}$).

2. Then, adding the $\log\det$ of the standardized Ψ and Θ as a penalty to the log likelihood will encourage their positive-definiteness.

Cooking new stuff

- (Exploratory) Structural Equation Modeling
- Hidden Markov Models
- (Multidimensional) Item Response Theory

Release date? We don't know

Download the **beta version** at https://github.com/Marcosjnez/latent

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